

Aproximação matemática de PI

```

#include "stdlib.h"
#include <stdio.h>
#include <time.h>

long long iters = 10000000000;

int main(int argc, char *argv[]) {
    double pi = 0;
    double sign = 1;

    struct timespec start, end;
    double elapsed;

    clock_gettime(CLOCK_MONOTONIC, &start);

    // Aproximação usando a sequencia de liebnitz
    for (size_t i = 1; i < iters; i += 2) {
        pi += (1 / (float)i) * sign;
        sign = -sign;
    }

    pi = 4 * pi;

    clock_gettime(CLOCK_MONOTONIC, &end);
    elapsed = (end.tv_sec - start.tv_sec) + (end.tv_nsec - start.tv_nsec) / 1e9;

    printf("Iters: %lld ", iters);
    printf("Elapsed wall time: %f seconds\n", elapsed);
    printf("Aprox: %f\n", pi);

    return 0;
}

// ❶ ~ /programming/hpc ❷ at 18:18:07 ))
// > ./a.out
// Iters: 1000000 Elapsed wall time: 0.002510 seconds
// Aprox: 3.141591
//
// ❶ ~ /programming/hpc ❷ at 18:18:08 ))
// > clang task3.c
//
// ❶ ~ /programming/hpc ❷ at 18:18:14 ))
// > ./a.out
// Iters: 1000000000 Elapsed wall time: 0.250464 seconds
// Aprox: 3.141593
//
// ❶ ~ /programming/hpc ❷ at 18:18:16 ))
// > clang task3.c
//
// ❶ ~ /programming/hpc ❷ at 18:18:22 ))
// > ./a.out
// Iters: 10000000000 Elapsed wall time: 24.917377 seconds
// Aprox: 3.141593

```